

VR-Numbers.

Operational Superstructures of Integers, Rationals, Reals, and Complex Numbers over the Natural Numbers of VR

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Abstract

Building on VR (Reznik, 2026), the axiom-free reconstruction of arithmetic from two primitive operations $\{\emptyset, t\}$, we develop **VR-Numbers** — a program of operational superstructures yielding $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ over VR’s natural numbers $\mathbb{N}$. The construction uses no set-theoretic (Kuratowski) ordered pairs — pairing is a syntactic *operation*, not an object; each numerical extension is a formal language of expressions with equivalence relations and operations. The entire system rests on the base operation $\emptyset$ alone — a nullary operation, carrying no ontology *of substance*; all numerical objects — including $\mathbb{N}$ — arise as operational constructions (terms over the operations) of progressively greater depth. Classical actual infinity is replaced by operational infinity: the induction principle is read as an operational principle, not as a postulate of completed infinite totalities. $\mathbb{C}_{VR}$ is the standard two-component construction over $\mathbb{R}_{VR}$ (a real and an imaginary part, joined by the postulate $i^2 = \ominus 1$), requiring no posited ordered pairs. The structures $\mathbb{Z}_{VR}, \mathbb{Q}_{VR}, \mathbb{C}_{VR}$ are isomorphic to their standard counterparts; $\mathbb{R}_{VR}$, built from Cauchy sequences with a *computable modulus of convergence*, is isomorphic to the field of computable real numbers — a countable subfield of classical $\mathbb{R}$ — reflecting the operational character of the construction. The Lean companion’s axiom profile is not inherited from VR’s empty `[]`: the superstructures rest on quotient types (`[propext, Quot.sound]`), and isomorphism bridges to the standard library (Lean Core’s `Rat`) pull `Classical.choice` (Part VIII).

Keywords: foundations of mathematics, operational extensions, operational infinity, integer/rational/real/complex construction, Cauchy sequences with computable modulus, computable analysis.

Version note (1.1.2). Scopes the “no ontology” language to *no ontology of substance* (§VI and the Summary): a wording correction — VR-Numbers ascribes no *substantial* primitive, yet does carry relational and structural commitments (the membership relation $\in$, the successor-identity of the $t^n(\emptyset)$), which are *tracked*, not denied; no construction, equivalence, operation, or theorem is altered. It otherwise inherits v1.1.1, which re-synchronised VR-Numbers with the parent work VR **v1.1.1**, which *removed* the propositional implication $\rightarrow$ (and the principles A1, A2 and the $F/\emptyset$ logical register): VR is now pure arithmetic on two operations $\{\emptyset, t\}$, generating no logic of its own. Accordingly Part I is restated for $\{\emptyset, t\}$, and **Part V no longer grounds the two-dimensionality of $\mathbb{C}$ in “the duality of A1”** — that grounding rested on the now-removed $F \rightarrow F/F \rightarrow \top$ axes and was an over-claim ($i^2 = \ominus 1$ is a separate postulate, as §V.4 notes, so A1 gave a coincidence of count, not a derivation). $\mathbb{C}_{VR}$ is presented as the standard two-component construction over $\mathbb{R}_{VR}$ with $i^2 = \ominus 1$. *No construction, equivalence, operation, or theorem is altered*; only the A1-grounding narrative is removed. (Earlier sync notes for v1.1.0 and 1.0.2–1.0.3 are superseded here.)

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Part I. Foundation: VR and $\mathbb{N}$ (axiom-free)

This work builds on the formal system VR (Reznik, 2026, *An Axiom-Free Reconstruction of Arithmetic from Operations Alone*; DOI 10.5281/zenodo.20633314). We recall its essentials in the form fixed by VR v1.1.1.

Primitive operations. VR is generated by **two** operations: the nullary base $\emptyset$ and the unary succession t . Every object is a *term* over this signature; nothing is posited beyond what these operations generate. (VR through v1.1.0 listed a third, a propositional implication $\rightarrow$; VR v1.1.1 removed it as a finite truth-function algebra used by no theorem — arithmetic, not logic. VR generates no logic; the logic used here is metatheoretic.)

No axioms of its own. What earlier versions called named principles are, under the operational reading, definitions or theorems over the freely generated term-algebra. (A1 and A2, which governed the removed propositional layer, no longer appear in VR v1.1.1.)

- **A3** is the *theorem* that $t(x)$ has the von Neumann content $x \in t(x)$ and $x \subseteq t(x)$; it follows from the definition of membership below. (t is the primitive succession; $t(x) = x \cup \{x\}$ is a gloss, with $\cup, \{\cdot\}$ abbreviations, not primitives.)
- **A4** (induction) is the *recursor* of the inductive term-algebra: the O_n exhaust the objects generated from $\emptyset$ by t .

Thus VR is axiom-free relative to the type-theoretic framework (the free term-algebra on $\{\emptyset, t\}$ and its recursor); what Peano postulates, VR derives.

Membership is defined. $\in$ is not a primitive but a relation defined by structural recursion: $z \in \emptyset := \perp$ and $z \in t(y) := (z \dot{=} y) \vee (z \in y)$ ($\perp, \vee$ metatheoretic; **mem** is **Prop**-valued), where $\dot{=}$ is structural term-identity (coinciding with Leibnizian equality below by theorem).

Natural numbers and equality. $O_0 = \emptyset$, $O_1 = t(\emptyset) = \{\emptyset\}$, $O_2 = \{\emptyset, \{\emptyset\}\}, \dots$; the natural number n is the term O_n . Equality is Leibnizian, $x = y :\Leftrightarrow \forall p : p(x) \leftrightarrow p(y)$, read schematically. VR is arithmetically equivalent to first-order PA and consistent relative to ZF (a metatheoretic corollary).

Operational status. There is no ontological primitive: $\emptyset$ is itself a nullary operation, characterised solely by $\forall x \neg(x \in \emptyset)$. All other objects, the naturals included, are terms over the operations: O_n is not an independent existing thing but the result of n applications of t to $\emptyset$. The content of n is its operational history. This operational position is the foundation of the present work: nothing simply “exists” — objects are terms over operations, of progressively greater depth.

Notation. Expression builders: $a \ominus b$ (integer), $a \oslash b$ (fraction), $a \oplus b \cdot i$ (complex). Operations are written with the symbols $\oplus, \otimes, \boxminus$ (integers), $\boxplus, \boxtimes, \boxminus_{\mathbb{Q}}, \oslash_{\mathbb{Q}}$ (rationals), and a subscript naming the system where needed ($\oplus_{\mathbb{R}}, \otimes_{\mathbb{R}}, \dots$); per §VI.3 every such symbol is read operationally.

Part II. Integers $\mathbb{Z}$ as an operational superstructure

II.1 Language of subtraction. Extend VR by a binary symbol $\ominus$. An expression $a \ominus b$ ($a, b \in \mathbb{N}$) records the operation “subtract b from a ”. It is not a set-theoretic (Kuratowski) pair-object but a formal record of an operation: $\ominus$ is a syntactic pairing *operation*, and $a \ominus b$ a term (a doing), not an object.

II.2 Equivalence. $a \ominus b \approx c \ominus d \Leftrightarrow a + d = b + c$ (using only $\mathbb{N}$ -addition). Reflexive, symmetric, transitive (by commutativity/associativity of $+$, and the right-cancellation law $a + c = b + c \Rightarrow a = b$ — see §VIII.1).

II.3 Operations. $(a \ominus b) \oplus (c \ominus d) := (a + c) \ominus (b + d)$; $(a \ominus b) \otimes (c \ominus d) := (a \times c + b \times d) \ominus (a \times d + b \times c)$; $\ominus(a \ominus b) := b \ominus a$; $(a \ominus b) \boxminus (c \ominus d) := (a + d) \ominus (b + c)$ (definable as $\oplus \ominus$). All well-defined on classes.

II.4 Canonical form. If $a \geq b$: $(a - b) \ominus 0$; if $a < b$: $0 \ominus (b - a)$.

II.5 Order. $[a \ominus b]$ is *positive* iff $a > b$ in $\mathbb{N}$ (i.e. $a = b + O_k$ for some $O_k \neq O_0$); and $[x] > [y] :\Leftrightarrow [x] \boxminus [y]$ is positive. This makes $\mathbb{Z}_{VR}$ a (decidably) ordered ring; positivity is decidable via the canonical form. (Order is inherited below by $\mathbb{Q}_{VR}$ and $\mathbb{R}_{VR}$; the symbols $\varepsilon > 0$, $|\cdot|$ used later refer to this order.)

II.6 Embedding $\mathbb{N} \rightarrow \mathbb{Z}_{VR}$. $n \mapsto [n \ominus 0]$.

Theorem 1. $\mathbb{Z}_{VR} = (\{[a \ominus b]\}, \oplus, \otimes, \boxminus, >)$ is an ordered commutative ring with unit, isomorphic to $\mathbb{Z}$.

Proof sketch. $\varphi([a \ominus b]) = a - b$ is well-defined on classes, respects operations and order, and is bijective.

Part III. Rationals $\mathbb{Q}$ as an operational superstructure

III.1 Language of division. A symbol $\oslash$; an expression $a \oslash b$ ($a, b \in \mathbb{Z}_{VR}$, $b \neq 0_{\mathbb{Z}}$) records “divide a by b ”. The restriction $b \neq 0_{\mathbb{Z}}$ is syntactic, decidable via the canonical form.

III.2 Equivalence. $a \oslash b \approx_{\mathbb{Q}} c \oslash d \Leftrightarrow a \otimes d = b \otimes c$ (integral-domain cancellation on $\mathbb{Z}_{VR}$; see §VIII.3).

III.3 Operations. $(a \oslash b) \boxplus (c \oslash d) := (a \otimes d \oplus b \otimes c) \oslash (b \otimes d)$; $(a \oslash b) \boxtimes (c \oslash d) := (a \otimes c) \oslash (b \otimes d)$; $(a \oslash b) \boxminus_{\mathbb{Q}} (c \oslash d) := (a \otimes d \boxminus b \otimes c) \oslash (b \otimes d)$; $(a \oslash b) \oslash_{\mathbb{Q}} (c \oslash d) := (a \otimes d) \oslash (b \otimes c)$, provided $c \neq 0_{\mathbb{Z}}$.

III.4 Canonical form. Make the denominator positive (in the $\mathbb{Z}_{VR}$ -order, §II.5) and reduce to lowest terms.

III.5 Order and embedding. $a \oslash b$ (denominator made positive) is positive iff a is positive in $\mathbb{Z}_{VR}$; this orders $\mathbb{Q}_{VR}$. Embedding $\mathbb{Z}_{VR} \rightarrow \mathbb{Q}_{VR}$: $a \mapsto [a \oslash 1_{\mathbb{Z}}]$.

Theorem 2. $\mathbb{Q}_{VR}$ with $\boxplus, \boxtimes, \boxminus_{\mathbb{Q}}, \oslash_{\mathbb{Q}}, >$ is an ordered field, isomorphic to $\mathbb{Q}$.

Part IV. Real numbers $\mathbb{R}$ via Cauchy sequences (computable modulus)

Real numbers require infinite processes, captured operationally as finite descriptions of them.

IV.1 Functions as operational rules. A function $a : \mathbb{N} \rightarrow \mathbb{Q}_{VR}$ is an operational rule: a finite description (algorithm) producing $a(n) \in \mathbb{Q}_{VR}$ for each n . Functions are not sets of ordered pairs but operational primitives. Each is captured by its finite description; the collection of such descriptions is operationally countable. Operationally, a is a partial recursive (Church–Turing, equivalently Type-2 computable) function (Weihrauch, 2000); the construction coincides with the standard development of computable reals via computable Cauchy sequences (Pour-El & Richards, 1989) — a natural operational route to the computable reals, the one adopted here. (Dedekind cuts with a computable predicate, nested intervals, or signed-digit representations would serve equally: the operational position dictates the *goal* — the computable reals — not a unique method.)

IV.2 Fundamental (Cauchy) sequences, with a computable modulus. A function $a : \mathbb{N} \rightarrow \mathbb{Q}_{VR}$ is *fundamental* if

$$\forall \varepsilon \in \mathbb{Q}_{VR}, \varepsilon > 0, \exists N \in \mathbb{N} : \forall m, n \geq N : |a(m) \boxminus_{\mathbb{Q}} a(n)| < \varepsilon.$$

Crucially, the witness N must itself be *operational*: a **modulus of convergence** $N(\varepsilon)$ given by a finite description (an algorithm computing N from ε). This is the operational reading of $\exists N$ (§VI.3) and is not an extra hypothesis but the content of “operational rule”. Without a computable modulus a (computable) *Specker sequence* — classically Cauchy but with a non-computable limit — would qualify, and its class would fall outside the computable reals, breaking §IV.7. With the modulus computable, the limit is computable (Pour-El & Richards). The Lean realisation cannot express this restriction in a single type and proves the wider classical statement; see §VIII.6.

IV.3 Equivalence. $a \approx_C b \Leftrightarrow \forall \varepsilon > 0, \exists N : \forall n \geq N : |a(n) \ominus_{\mathbb{Q}} b(n)| < \varepsilon$ (with a computable modulus, as in §IV.2).

IV.4 The construction. $\mathbb{R}_{VR}$ is the set of $\approx_C$ -classes of fundamental sequences.

IV.5 Operations. Componentwise on representatives: $[a] \oplus_{\mathbb{R}} [b] := [n \mapsto a(n) \boxplus b(n)]$; $[a] \otimes_{\mathbb{R}} [b] := [n \mapsto a(n) \boxtimes b(n)]$; $[a] \ominus_{\mathbb{R}} [b] := [n \mapsto a(n) \ominus_{\mathbb{Q}} b(n)]$. *Division* $[a] \oslash_{\mathbb{R}} [b]$, for $[b] \neq 0$, requires a representative **bounded away from zero** (apartness): since $[b] \neq 0$ there are k and a positive rational δ with $|b(n)| > \delta$ for all $n \geq k$; choosing such a representative, $[a] \oslash_{\mathbb{R}} [b] := [n \mapsto a(n) \oslash_{\mathbb{Q}} b(n)]$ (for $n \geq k$) is well-defined. Apartness from 0, not mere $[b] \neq 0$, is what makes componentwise division total on a chosen representative.

IV.6 Embedding $\mathbb{Q} \rightarrow \mathbb{R}_{VR}$. $q \mapsto [n \mapsto q]$ (constant sequence).

IV.7 Isomorphism theorem.

Theorem 3. $\mathbb{R}_{VR}$ is a real closed ordered field, isomorphic to the field of computable real numbers — a countable subfield of the classical real line $\mathbb{R}$.

What Lean certifies. The machine-checked isomorphism is with the *full classical* $\mathbb{R}$ (the Lean type $\mathbb{N} \rightarrow \mathbb{Q}_{VR}$ contains every total function); the restriction to the *computable* reals stated in the theorem is metatheoretic — see §VIII.6. The gap is deliberate and worth naming: the position’s object (a countable, *computable* $\mathbb{R}_{VR}$) and the machine-certified object (the classical, uncountable $\mathbb{R}$) are not the same — the formalisation verifies *more* than the operational thesis claims; and since $\mathbb{C}_{VR}$ is built over $\mathbb{R}_{VR}$, the same gap carries to $\mathbb{C}$. Real-closedness — that roots of polynomials with computable coefficients are computable, notwithstanding the undecidability of the sign of a computable real — is the strongest algebraic claim here (Pour-El & Richards, 1989).

Remark 1. $\mathbb{R}_{VR}$ is not isomorphic to the entire classical $\mathbb{R}$: the latter has the cardinality of the continuum, whereas $\mathbb{R}_{VR}$, consisting of operational rules each given by a finite description with a computable modulus, is countable. Concretely, $\mathbb{R}_{VR}$ captures exactly the reals *admitting an algorithm that produces arbitrarily close rational approximations* (the computable reals) — the algebraic numbers, π , e , and so on. A real merely *definable* by a finite description need not be computable (e.g. Chaitin’s Ω , or the halting real) and is not among them; non-computable reals of classical $\mathbb{R}$ have no place here, corresponding to no algorithm. This aligns VR-Numbers with the constructive tradition (Bishop, 1967; Martin-Löf, 1984).

Remark 2 (relation to the continuum). This $\mathbb{R}_{VR}$ is the *Cauchy/computable* operational real — countable. It is a distinct construction from the choice-sequence (Brouwer) continuum pursued elsewhere in the cycle, which is non-enumerable; the two are different operational readings of “the reals”, not the same object.

IV.8 Operational character. A real in VR is, formally, an algorithm: a finite description specifying how to compute arbitrarily close rational approximations (e.g. $\sqrt{2}$ by Newton’s method).

Part V. Complex numbers $\mathbb{C}$

In standard mathematics $\mathbb{C}$ is built as ordered pairs of reals with a special multiplication. We build it without posited ordered pairs — as a two-component structure over $\mathbb{R}_{VR}$ (a real and an imaginary part), with the algebraic coupling $i^2 = \ominus 1$.

V.1 Two components. A complex value carries two real components: a *real* part (along which $\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R}$ unfolds) and an *imaginary* part, where expressions $b \cdot i$ live; i is not a number but a syntactic marker of which component a value belongs to. (Earlier versions ascribed the two components to a “duality of A1” — the implications $F \rightarrow F/F \rightarrow T$; that grounding is dropped in v1.1.1 with the removal of the propositional layer. The two-dimensionality is simply the construction’s two components, not a consequence of logic.)

V.2 The joining rule. To make $\mathbb{C}$ one structure rather than two disjoint copies of $\mathbb{R}$, the two components are coupled by $i \otimes i := \ominus 1$ (i.e. -1): applying the imaginary marker twice equals negation along the real axis. This joining rule is an additional postulate.

V.3 The construction. A complex number is a formal expression $a \oplus b \cdot i$ ($a, b \in \mathbb{R}_{VR}$) — a syntactic record of two real components, not a set-theoretic ordered pair (pairing here is a syntactic operation). Equivalence: $a \oplus b \cdot i \approx c \oplus d \cdot i \Leftrightarrow a = c$ and $b = d$ in $\mathbb{R}_{VR}$. Operations: $(a \oplus b \cdot i) \oplus_{\mathbb{C}} (c \oplus d \cdot i) := (a \oplus_{\mathbb{R}} c) \oplus (b \oplus_{\mathbb{R}} d) \cdot i$; $(a \oplus b \cdot i) \otimes_{\mathbb{C}} (c \oplus d \cdot i) := (a \otimes_{\mathbb{R}} c \boxminus_{\mathbb{R}} b \otimes_{\mathbb{R}} d) \oplus (a \otimes_{\mathbb{R}} d \oplus_{\mathbb{R}} b \otimes_{\mathbb{R}} c) \cdot i$; conjugation $a \oplus (\ominus b) \cdot i$; modulus $\sqrt{a \otimes_{\mathbb{R}} a \oplus_{\mathbb{R}} b \otimes_{\mathbb{R}} b}$. Embedding $\mathbb{R} \rightarrow \mathbb{C}_{VR}$: $a \mapsto a \oplus 0 \cdot i$.

Theorem 4. $\mathbb{C}_{VR}$ is a field, isomorphic to the standard $\mathbb{C}$ (with $\mathbb{R}$ replaced by $\mathbb{R}_{VR}$; the result is the field of computable complex numbers).

V.4 On two-dimensionality. The two-dimensionality of $\mathbb{C}$ is the construction’s two components together with the postulate $i^2 = \ominus 1$; it is not derived from anything more primitive. (Earlier versions claimed a “logical origin” in the duality of A1; this is withdrawn — it rested on the now-removed $F \rightarrow F/F \rightarrow T$ axes, and even then $i^2 = \ominus 1$ was admittedly a separate postulate, so A1 supplied a coincidence of count, not a derivation.) The theorems of Frobenius (1878) and Hurwitz (1898) — no further normed division algebra without weakening commutativity or associativity — delimit the extension; whether $\mathbb{H}, \mathbb{O}$ admit operational constructions is open (Part VII).

Part VI. The operational position of VR-Numbers

VR-Numbers ascribes no ontology *of substance*: there is no ontological primitive — $\emptyset$ is a nullary operation, characterised solely by $\forall x \neg(x \in \emptyset)$. All numbers are operational constructions (terms over the operations) of progressively greater depth. This is not the claim that the system ascribes *no ontology at all*: it carries relational and structural commitments — the membership relation $\in$ used to characterise $\emptyset$, and the successor-identity of the $t^n(\emptyset)$ — which are *tracked*, not denied. What is absent is any *substantial* primitive: an object that *is* prior to operations.

VI.1 No ontological primitive. $\emptyset$ carries no ontological content — its entire characterisation is $\forall x \neg(x \in \emptyset)$, a fact about $\in$, not a description of contents. Nothing is “prior to operations”; $\emptyset$ is itself an operation. t is a rule, not an object; the $O_n = t^n(\emptyset)$ are records of successive applications, and that is the totality of what they are.

VI.2 Depths of operationality.

System	Operational character
$\emptyset$	The nullary base operation — depth zero of the construction (no ontology of substance)
$\mathbb{N}$	Direct applications of t to $\emptyset$: $O_n = t^n(\emptyset)$
$\mathbb{Z}$	Finite syntactic expressions $a \ominus b$ over $\mathbb{N}$
$\mathbb{Q}$	Finite syntactic expressions $a \oslash b$ over $\mathbb{Z}$
$\mathbb{R}$	Algorithms (computable modulus) producing rational sequences
$\mathbb{C}$	Two real components $a \oplus b \cdot i$ over $\mathbb{R}_{VR}$, joined by $i^2 = \ominus 1$

VI.3 Operational infinity. The induction principle A4 is conventionally read as postulating actual infinity — the completed totality of all naturals as an object. We reject this reading: A4 asserts no infinite object, but the operational principle that any property holding for $\emptyset$ and preserved under t holds for every result of iterated t . The quantifier “for all n ” is operational — universal applicability of an operation, not membership in a completed set. We call this *operational infinity*.

Operational and actual infinity **agree on arithmetical consequences**: via the equivalence $VR \leftrightarrow$ first-order PA, every *arithmetical* theorem provable with actual infinity is provable here with operational infinity (and conversely). They differ in what they posit as existing: actual infinity asserts an infinite totality as an object; operational infinity asserts only that an unbounded operation is available. *Beyond arithmetic the two diverge*: classical mathematics with actual infinity proves, e.g., the uncountability of $\mathbb{R}$ and the existence of non-computable reals — theorems with no counterpart here, since $\mathbb{R}_{VR}$ is countable (§IV.7). The equivalence is therefore at the level of arithmetic, not of all classical mathematics; the operational position does not reconstruct theorems essentially depending on completed uncountable totalities. This is a feature: it bypasses the Aristotelian/intuitionist objection to completed infinities without losing arithmetic.

All references to “infinite” objects ($\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$) are read operationally: none is an object; each is a domain of applicability for operations over $\emptyset$. The set-theoretic vocabulary is retained for conformity with standard practice, with the operational reading understood (the operational reconstruction of that vocabulary is the subject of VR-Sets).

VI.4 The gradient of operationality. Each successive system is a deeper operational construction over $\emptyset$: naturals (direct t -iteration), integers/rationals (finite syntax with equivalence), reals (finite descriptions of infinite processes, with computable modulus), complex (a second real component, joined by $i^2 = \ominus 1$). The gradient mirrors the historical introduction of each extension and the conceptual difficulty that accompanied it.

VI.5 What does not exist as an object. Ordered pairs (replaced by syntactic juxtaposition $a \ominus b, a \oslash b, a \oplus b \cdot i$), negative numbers, fractions, reals, complex numbers — none is introduced as a primitive object; each is a notation for an operational class or expression. *Nothing is — all is doing.*

VI.6 Consistency relative to ZF. $\mathbb{Z}_{VR}, \mathbb{Q}_{VR}, \mathbb{C}_{VR}$ are isomorphic to their standard counterparts; the computable reals form a definable subfield of $\mathbb{R}$ in ZF (Pour-El & Richards, 1989) and $\mathbb{R}_{VR}$ is isomorphic to it. With the standard ZF-models for $\mathbb{Z}, \mathbb{Q}, \mathbb{C}$ and the consistency of VR relative to ZF, VR-Numbers is consistent relative to ZF (a metatheoretic corollary).

Part VII. Open questions

- (1) **Type theory.** The operational view of functions ($\mathbb{R}$) and the marker i ($\mathbb{C}$) suit Martin-Löf-style type theories; the Lean 4 formalisation (Part VIII) is a first step.
- (2) **Beyond $\mathbb{C}$.** Whether $\mathbb{H}, \mathbb{O}$ (Frobenius 1878, Hurwitz 1898) admit analogous operational constructions.
- (3) **VR-Sets.** The operational reconstruction of set theory — membership as “ x was used in the construction of y ”, unifying well-founded (ZFC) and non-well-founded (ZFA) modes — is now available (Brouwer edition, DOI 10.5281/zenodo.20643407).
- (4) **Computational realization.** Each level admits direct realization for exact-arithmetic computation.

Part VIII. Notes from the Lean 4 formalisation

Parts II–V are formalised in Lean 4 (Reznik, 2026, DOI 10.5281/zenodo.20352057; on Part I’s DOI 10.5281/zenodo.20324240), ≈ 3600 lines (Integers/Rationals/Reals/Complex), no `sorry`/`admit`, axiom dependencies documented. Eight observations, in three groups.

VIII.1 Hidden lemma (§II.2): cancellation of addition. “Direct verification” splits into unequal steps; transitivity needs right-cancellation $a + c = b + c \Rightarrow a = b$, proved by induction on c using successor injectivity (P4), not by T1–T4 rewriting.

VIII.2 Hidden lemma (§II.3): left distributivity. $(a + b) \times c = a \times c + b \times c$ is not in T1–T4 (commutativity of $\times$ is not assumed) and is proved by induction on the right argument; the formalisation makes this restraint visible.

VIII.3 Hidden lemma (§III.2): multiplicative cancellation. Transitivity of $\approx_{\mathbb{Q}}$ needs $xz = yz \Rightarrow x = y$ ($z \neq 0$); proved constructively via `Int.natAbs`, avoiding `mathlib`’s general `IntegralDomain` (which would pull `Classical.choice`). Integral-domain cancellation on $\mathbb{Z}$ is constructive; the choice-dependence is infrastructural, not intrinsic.

VIII.4 Boundary: Classical.choice via the $\mathbb{Q}$ -isomorphism. $\mathbb{Q}_{VR}$ is fully constructive (`Quot.sound`, `propext` only). The $\mathbb{Q}_{VR}$ construction itself — the expressions `NonZeroRatExpr`, the equivalence `ratEq`, and the operations on its classes — is *also* constructive (`[propext, Quot.sound]`, no `Classical.choice`). The choice enters only through the *isomorphism* $\mathbb{Q}_{VR} \cong \mathbb{Q}$ (Theorem 2), which maps classes to Lean Core’s `Rat` (via `Rat.divInt`), and `Rat.add`/`Rat.mul` normalise through `Nat.gcd` (`[propext, Classical.choice, Quot.sound]`). So the field theorems proved by transport along that bridge inherit the choice — a feature of Lean’s `Rat` representation, not of the rationals (a non-normalising `Rat` would stay constructive), and *not* of the operational construction of Part III, which is choice-free.

VIII.4a Canonical realisation: the choice-free operational line is the primary one. The isomorphism $\mathbb{Q}_{VR} \cong \mathbb{Q}$ of §VIII.4 is an *interoperability* bridge (it reuses Lean Core’s field theorems), not the canonical construction of the VR line, and it is the *only* place `Classical.choice` appears in the number tower. The canonical operational number line is realised *end to end* below the choice

floor by the cycle’s `Continuum` module, independently of that bridge: `Continuum.Rational`’s `Qop` is a choice-free field ($\mathbb{Q}$ operationally, profile `[propext, Quot.sound]`, with a total inverse got from a zero-test, no `Decidable` instance); `Continuum.GaussianRational`’s `GaussQ` carries the operational Gaussian rationals (a $\mathbb{C}$ layer) inheriting that profile; and `Continuum.Real` / `Continuum.UnitInterval` build an operational $\mathbb{R}$ by hand from $\mathbb{Z}$, also below the floor. These are not asserted but machine-verified: `Continuum.Spectrum` discharges guards of the form `#assert_not_depends_on Qop.mul_inv_cancel on Classical.choice`. Choice re-enters only on the optional mathlib-facing *labels* — the `Field/ratCast/ofRat` casts and the $\cong$ -mathlib theorems — never on the operations themselves; the *doing* is choice-free, only the Tier-3 interop label is not. So the claim “the VR number line is choice-free, machine-checked” stands under `#print axioms` on the canonical objects; the choice recorded in §VIII.4 is a property of the reused Lean-Core `Rat` representation across the interop bridge, not of the operational line VR constructs.

VIII.5 Boundary: the quotient becomes vacuous at $\mathbb{C}$. $\mathbb{Z}_{VR}, \mathbb{Q}_{VR}, \mathbb{R}_{VR}$ share the pattern record/equivalence/quotient. At $\mathbb{C}_{VR}$ the equivalence is componentwise equality (§V.3), coinciding with built-in equality on a two-field structure: the quotient is vacuous, $\mathbb{C}_{VR}$ is the first type realised without `Quotient`. The depth at $\mathbb{C}$ is carried by the two-dimensionality (its two components, $i^2 = \ominus 1$), not by an equivalence.

VIII.6 Boundary: inexpressibility of computability on $\mathbb{R}_{VR}$. §IV.1 stipulates functions $\mathbb{N} \rightarrow \mathbb{Q}_{VR}$ are operational rules and §IV.7 concludes $\mathbb{R}_{VR} \cong$ the computable reals. But the Lean type $\mathbb{N} \rightarrow \mathbb{Q}_{VR}$ contains *every* total function of that signature (cardinality of the continuum), so the formalisation proves an isomorphism with the *full classical* $\mathbb{R}$, not the computable subfield: Lean 4 has no type-level predicate separating computable from non-computable functions, so the operational restriction (§IV.1–IV.2, including the computable modulus) is metatheoretic relative to Lean. Recommended reading within this cycle: the formalisation faithfully realises the Cauchy construction (§IV.1–IV.6) and additionally proves a structural isomorphism with classical $\mathbb{R}$; the restriction to computable reals (§IV.2, §IV.7) is preserved as metatheoretic commentary. VR-Forms provides the formal register in which the operational/non-operational relation is addressed directly.

VIII.7 Pattern: tactics cleaner than named lemmas. `mul_zero`, `mul_right_cancel0`, `linarith`, `linear_combination` carry latent `Classical.choice` via typeclass generality on concrete numeric types; `ring`, `omega`, manual identities discharge the same goals Classical-free. Prefer tactics to named algebraic lemmas for axiom-minimality on concrete types.

VIII.8 Pattern: change vs simp. At three points in $\mathbb{R}_{VR}$, `simp` drove goals into incompatible forms (`Rat.blt`; `abs_neg` ring-normalisation; `Real.mk_eq`); replacing it with `change` (definitional restatement, no reduction) kept syntactic control. The lighter tactic is preferable where it suffices.

Summary

VR-Numbers extends VR with operational superstructures yielding $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$. Three theses: (1) *No ontology of substance* — $\emptyset$ is a nullary operation; all objects are terms over operations; no set-theoretic ordered pairs are primitive — pairing is a syntactic operation (a binary constructor), not an object. (2) *Actual infinity is replaced by operational infinity* — induction is universal applicability of an operation, agreeing with actual infinity on arithmetical consequences (via $VR \leftrightarrow PA$) while positing no completed totality. (3) *$\mathbb{C}$ is built without posited ordered pairs* — as two real components

$a \oplus b \cdot i$ over $\mathbb{R}_{VR}$, coupled by the postulate $i^2 = \ominus 1$ (its two-dimensionality is the construction's, not a consequence of logic). $\mathbb{Z}_{VR}, \mathbb{Q}_{VR}, \mathbb{C}_{VR}$ are isomorphic to their standard counterparts; $\mathbb{R}_{VR}$ (Cauchy sequences with a computable modulus) is isomorphic to the computable reals, a countable subfield of $\mathbb{R}$. Consistency relative to ZF follows metatheoretically. The position is minimal: *nothing is — all is doing*.

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